Dissertation Defense: Hybrid Neural Network Model Defenses

Eric Yocam

Location:
Madlabs Briefing Room
Dakota State University, Madison, SD

Date: Wednesday**,** March 25, 2026
Time: 4:30 PM – 5:30 PM

Acknowledgments

Dr. Varghese M. Vaidyan
Committee Chair

Dr. Austin O'Brien
Committee Member

Dr. Gurcan Comert
Committee Member

I would like to extend my deepest gratitude to my committee for their guidance and support throughout this journey.

Agenda

- **Introduction**
- Literature Review
- Methodology
- Results
- Discussion
- Conclusion

Eric Yocam Bio

Bio-Sketch	
Teaching – Cybersecurity at Cal Poly State University, Dominican Univ. of California & Other Institutions	Professional – Startup Companies, Apple, HP, Microsoft, T-Mobile
Inventor – 20+ Issued Patents	Sr. Member IEEE & Contributor to ACM, ISC2, ISACA, EC-Council, ABET
Hard Sci-Fi Writer & Life-long Learner	X-Force Fellowship, Teaching Excellence Award Univ. of Washington, ACM Featured Reviewer

15 years teaching online/in person and **25+** years as a professional

Publication Activity Summary

Type	Count
Published	8
In Production	1
Under Peer Review	6
In Development	3

Total of **8** publications + **10** related works

Publications Resulting from this Research

No.	Progress	Date	Venue	Title
1	**In Production**	Apr. 2026	Book Chapter - Springer Quantum Science and Technology Series	Quantum Adversarial Machine Learning and Defense Strategies: Challenges and Opportunities
2	**Under Peer Review**	Jan. 2026	Transportation Research Interdisciplinary Perspectives (Elsevier)	Architectural Robustness in Lane Detection Systems: Geometric Constraints and Adversarial Stability
3	**Under Peer Review**	Dec. 2025	IEEE Transactions on Emerging Topics in Computational Intelligence	Protection of UAV-Based Crop Row Detection Against Adversarial Attacks
4	**Under Peer Review**	Aug. 2025	IEEE Transactions on Intelligent Transportation Systems	Reflectivity-Driven Traffic Sign Safety with Synthetic Data and Defense Models
5	**Published**	Feb. 2024	IEEE International Conference on Computing, Networking and Communications Workshop	Security Dataset Augmentation Invariance and Distribution Independence
6	**Under Peer Review**	Mar. 2026	IEEE Transactions on Dependable and Secure Computing	Hardening Black-Box Neural Networks Against Adversarial Attacks Through Surrogate-Based Defense Transfer

Additional Research Contributions from the PhD (1 of 2)

No.	Progress	Date	Venue	Title
1	**Published**	Mar. 2026	IEEE Access	A Dynamic Quantum-Resistant Selective Encryption Approach for Agricultural Sensors with Limited Resources
2	**Published**	Aug. 2024	IEEE Transactions on Emerging Topics in Computational Intelligence	Balancing Security and Correctness in Code Generation: An Empirical Study on Commercial Large Language Models
3	**Published**	2025	IEEE 15th Annual Computing and Communication Workshop and Conference	A Novel Approach to Quantum-Resistant Selective Encryption for Agricultural Sensors with Limited Resources
4	**Published**	2025	IEEE International Conference on Electro Information Technology	Multi-level Post Quantum Encryption for Images with Quantum Fourier Transform
5	**Under Peer Review**	Oct. 2025	IEEE Transactions on Emerging Topics in Computational Intelligence	Hybrid Quantum-Classical Path Optimization for Multi-Drone Gas Leak Detection

Additional Research Contributions from the PhD (2 of 2)

No.	Progress	Date	Venue	Title
1	**Published**	Mar. 2026	IEEE Access	A Novel Hybrid Quantum-Classical Path Optimization for Methane Detection Using Remote Quantum Intensity Prediction Models
2	**Published**	Oct. 2024	IEEE Transactions on Emerging Topics in Computational Intelligence	From LLMs to Randomness: Analyzing Program Input Efficacy through Resource Metrics and Language Embeddings
3	**Published**	Sep. 2024	IEEE Access	Evaluating Large Language Models for Enhanced Fuzzing: An Analysis Framework for LLM-Driven Seed Generation
4	**Under Peer Review**	2026	IEEE Transactions on Emerging Topics in Computational Intelligence	Cross-Domain Adversarial Robustness of Geometric Prior Neural Networks: Safety-Critical Deployment Guidelines

Additional Research Contributions from the PhD – In Development

No.	Progress	Date	Venue	Title
1	**Preprint**	2026	arXiv:2603.10195v1	Adaptive Activation Cancellation for Hallucination Mitigation in Large Language Models
2	**In Dev**	2026	IEEE Transactions on Emerging Topics in Computational Intelligence	Event-Driven Methane Leak Prediction with Quantum Spiking Neural Networks
3	**In Dev**	2026	IEEE Access	Semantic Multi-Tier Quantum-Resistant Encryption for Agricultural IoT with Hardware Validation

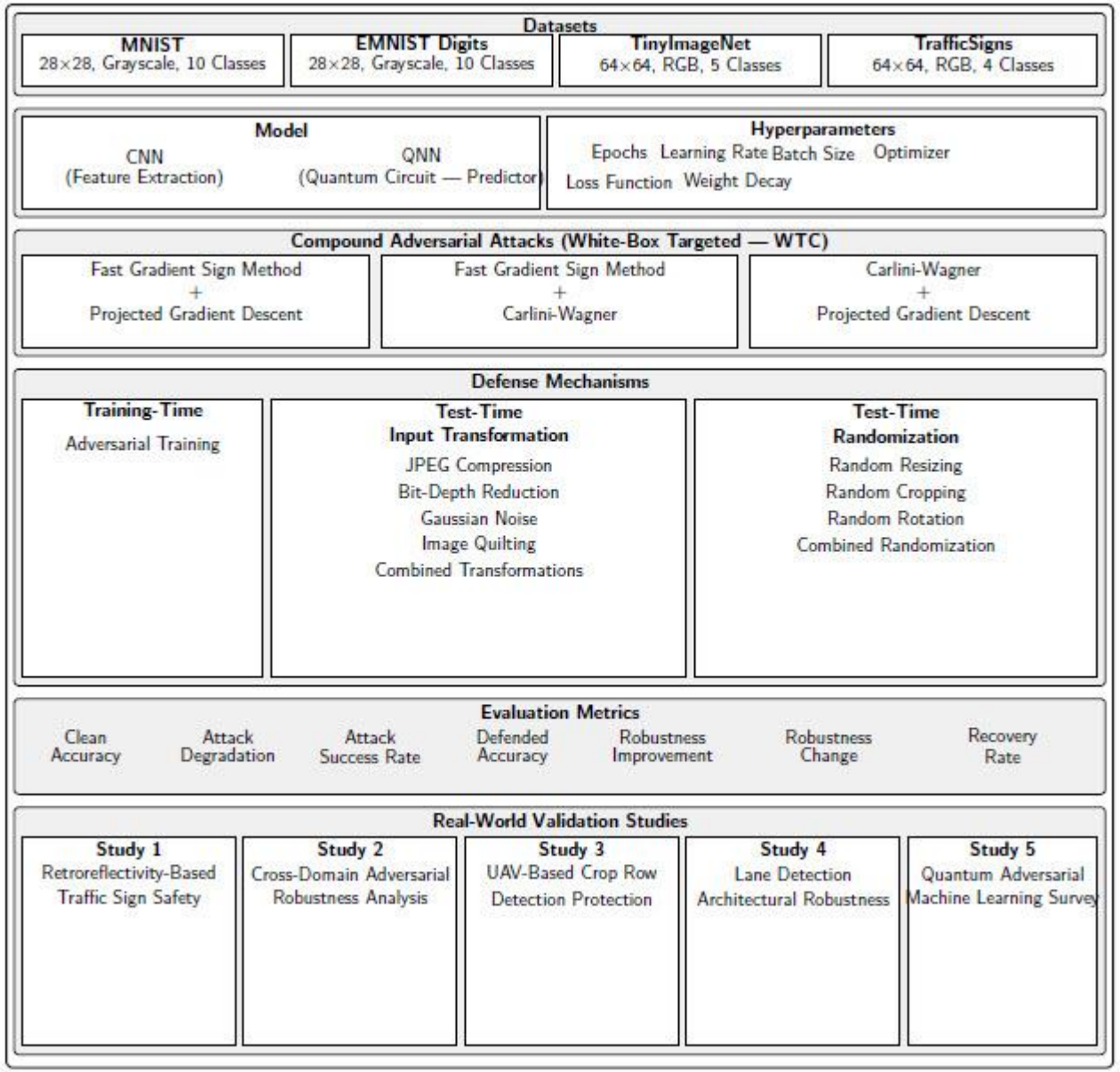
Overview (This Research)

Motivation

- HNNs lack compound defenses
- Prior work: single attacks only
- No systematic defense evaluation

Novelty

- First compound HNN evaluation
- First multi-defense comparison
- First dataset-dependent analysis



Contributions (This Research)

Compound Attack Evaluation

- First systematic study of WTC compound attacks on HNN models
- Established severity baseline: accuracy drops from 87.6% to 19.1% under compound attacks

Defense Mechanism Comparison

- First multi-defense systematic comparison on HNN models
- Adversarial training outperformed all test-time defenses by 2.3–2.8× across all datasets

Test-Time vs. Training-Time Framework

- Revealed deployment flexibility vs. robustness trade-off for HNN defenders
- Adversarial training: highest accuracy recovery but requires model retraining

Dataset-Dependent Defense Effectiveness

- HNN defenses achieved 13–18% relative robustness gains over classical CNN baselines
- Dataset-dependent resilience: 87% recovery on TrafficSigns vs. 28% on EMNIST Digits

Adversarial training is the only defense that consistently protects HNN models — but dataset complexity determines how much recovery you get.

Relevance to Cyber Operations

- **Bad actors** exploit white-box access to attack hybrid AI systems
- **HNN architectures** emerging in edge AI and defense platforms
- Identifies **adversarial weaknesses and benchmarks** compound defenses
- Supports **offensive and defensive cyber operations** against next-gen AI

Hybrid AI systems in critical infrastructure are under attack — and no defenses had been benchmarked.

Agenda

Introduction

Literature Review

Methodology

Results

Discussion

Conclusion

Literature Review

Adversarial Attacks

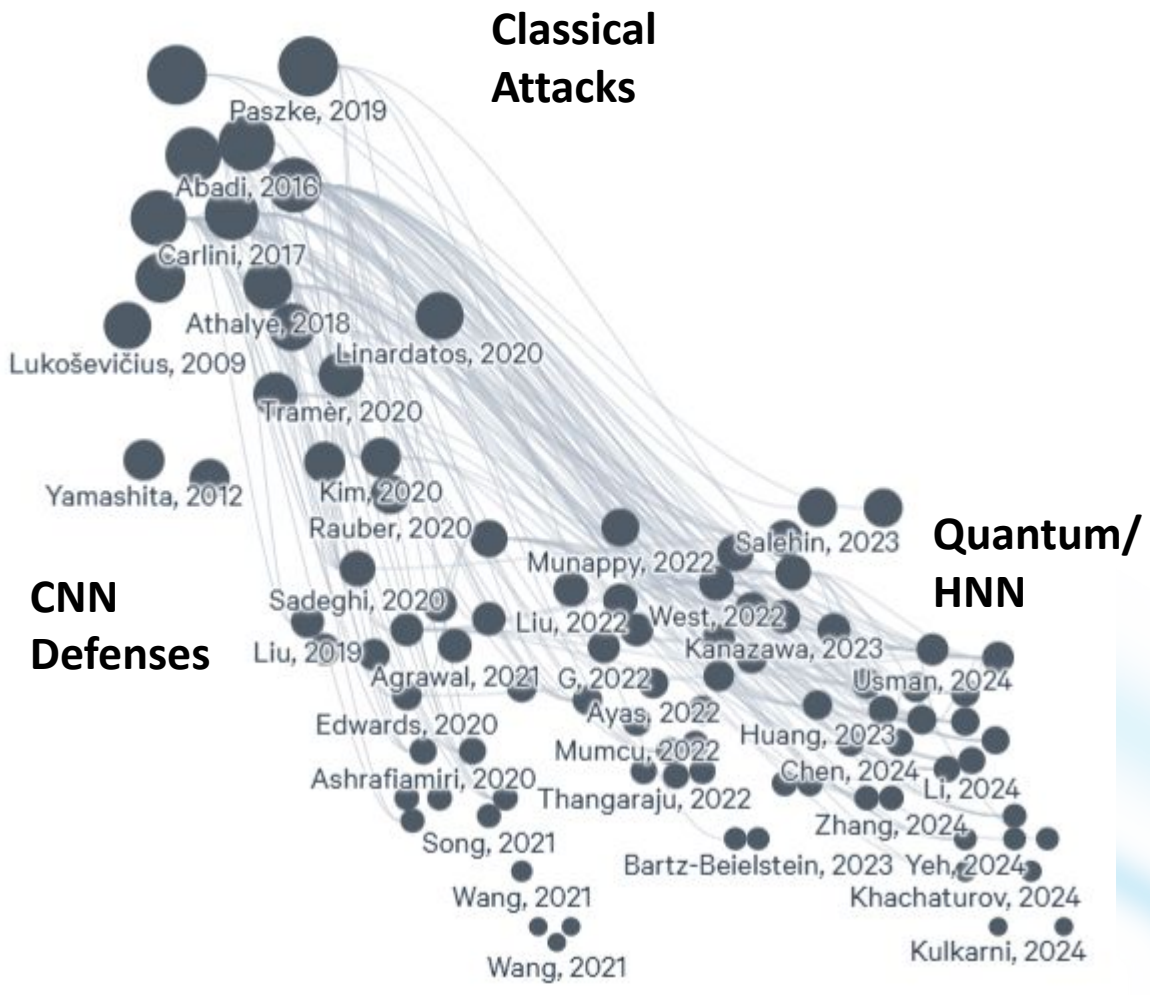
- Compound attacks (FGSM, PGD, CW) evaluated in isolation
- No compound attack framework exists for HNN models

Defense Mechanisms

- No systematic multi-defense comparison on hybrid architectures
- Test-time and training-time defenses not benchmarked together

Quantum/HNN Security

- Quantum-classical models treated as black boxes in security literature
- No adversarial defense framework exists for HNN architectures



Literature Map of Related HNN Papers

(Generated from: litmaps.com)

Research Gap (This Research)

- **No** compound attack framework existed for HNN models
- **No** systematic multi-defense comparison on HNN architectures
- **No** benchmark distinguishing test-time vs. training-time defenses on HNNs
- **No** analysis of dataset complexity effects on HNN defense effectiveness
- **No** validated methodology for HNN defense research using classical simulation

No prior work has systematically evaluated compound attacks or defense mechanisms against hybrid neural network models.

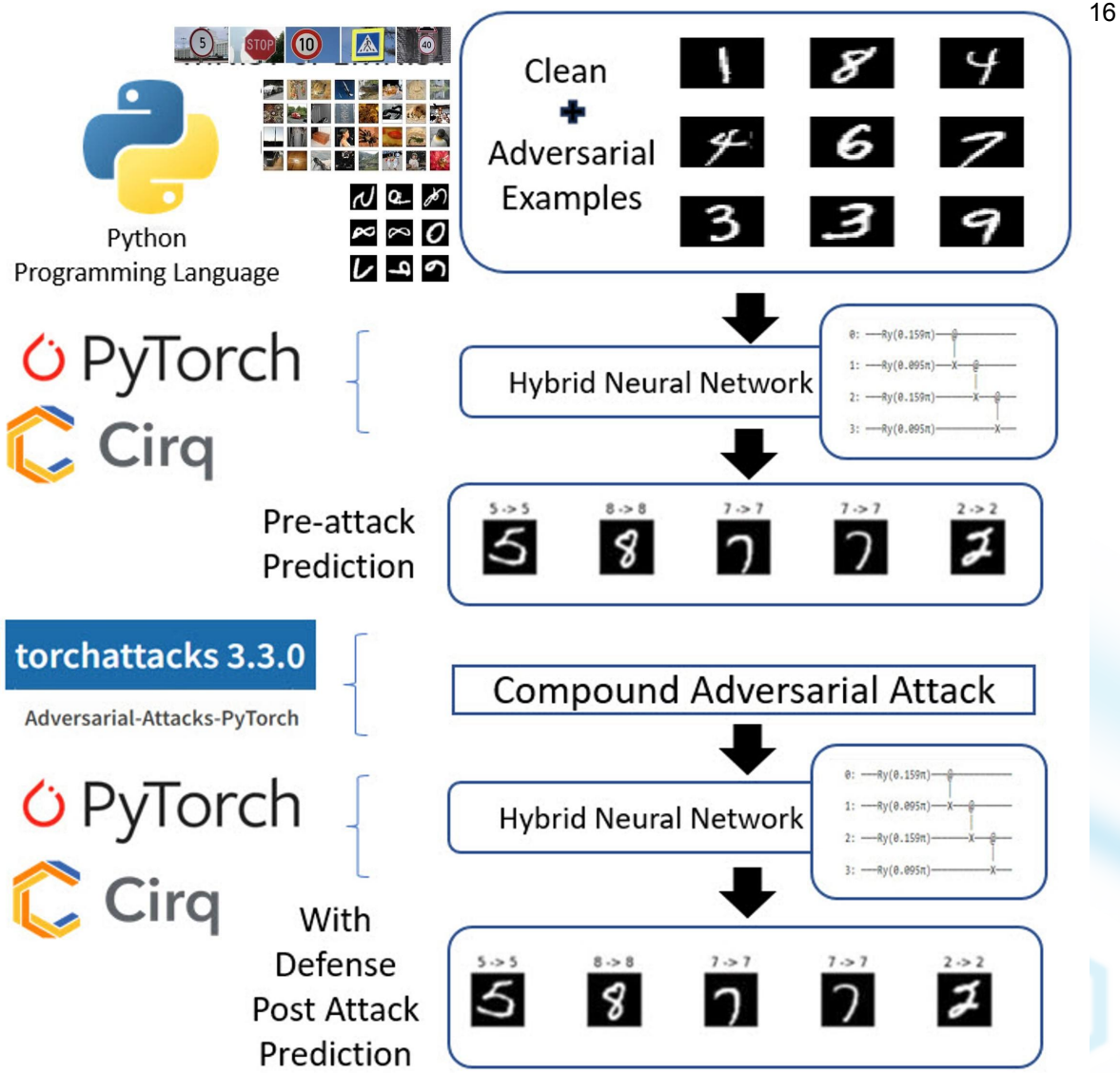
HNN Model Defense

Defense Strategies

- Test-time defenses offer flexibility without retraining
- Adversarial training provides strongest recovery

Evaluation Framework

- 120 conditions across 4 datasets and 3 attack combinations
- 7 metrics from clean accuracy to recovery rate



HNN Model Defense Process Context

Agenda

- Introduction
- Literature Review
- **Methodology**
- Results
- Discussion
- Conclusion

Methodology Goals

Research Design

- Demonstrate defense mechanisms protect HNN models against WTC adversarial attacks
- Compare input transformation, randomization, and adversarial training strategies

Defense Evaluation

- Identify optimal defense using prediction accuracy and robustness as performance metrics
- Evaluate FGSM+PGD, FGSM+CW, and PGD+CW across 4 datasets

Goal: Demonstrate that defense mechanisms protect an HNN model against WTC adversarial attacks

Methodology Overview

Experimental Design

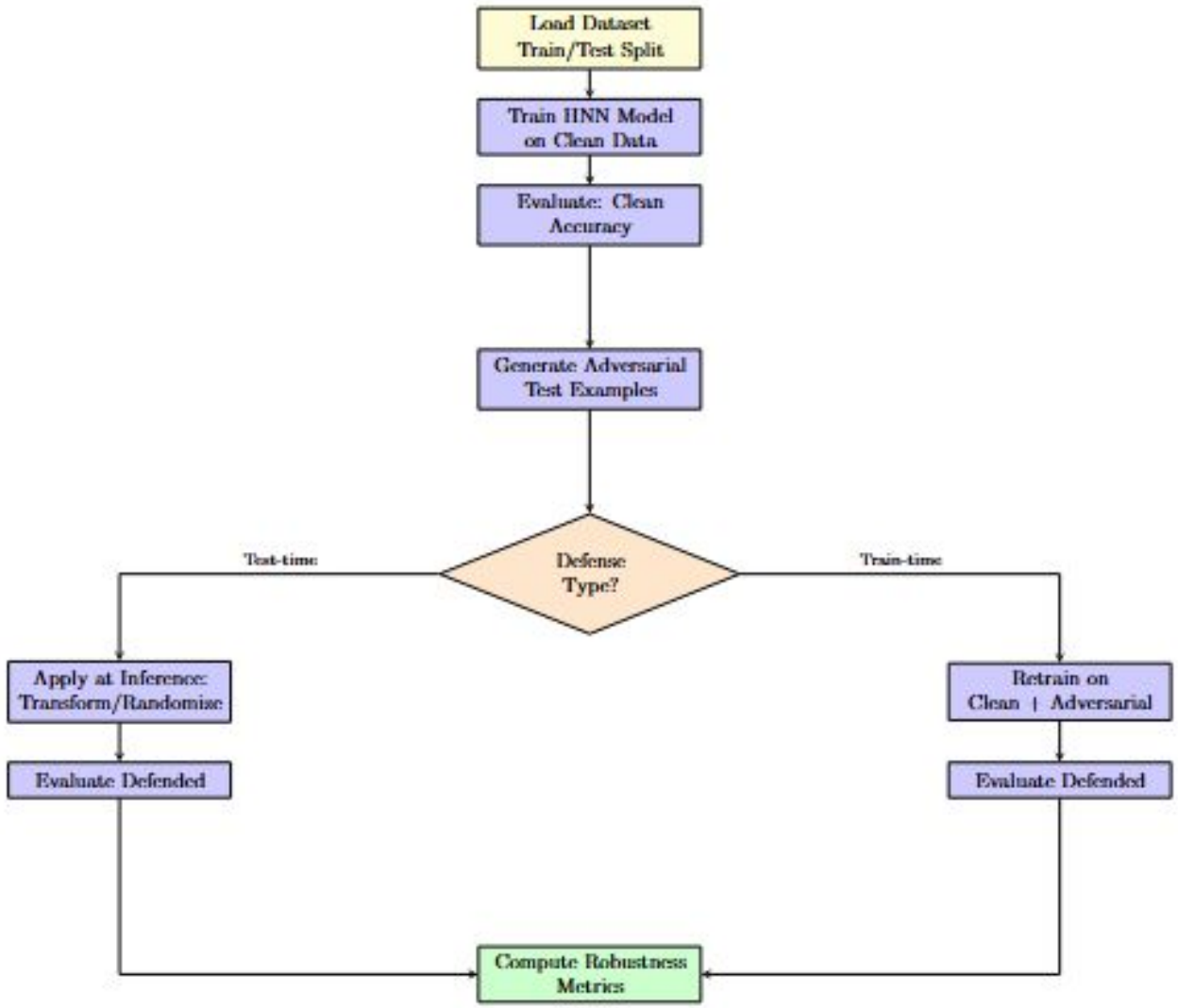
- 9-layer HNN with 4-qubit quantum circuit across 4 datasets
- 3 compound WTC attacks × 10 defenses = 120 conditions

Evaluation Approach

- Pre-attack baseline → compound attack → defense applied
- Compare test-time vs. training-time defense strategies

Measures

- Clean accuracy, attack degradation, defended accuracy
- Recovery rate, robustness improvement, defense rating



Measurement & Calculations

Attack Effectiveness

- Attack Degradation (Att Deg) = Clean Accuracy – Attacked Accuracy
- Attack Success Rate (Att SR) = percentage of clean accuracy lost

Defense Effectiveness

- Robustness Improvement (Rob Imp) = Defended Accuracy – Attacked Accuracy
- Robustness Change (Rob Change) = % improvement over attacked baseline
- Recovery Rate (Rec Rate) = % of lost accuracy recovered

Overall

Gap = Clean Accuracy – Defended Accuracy

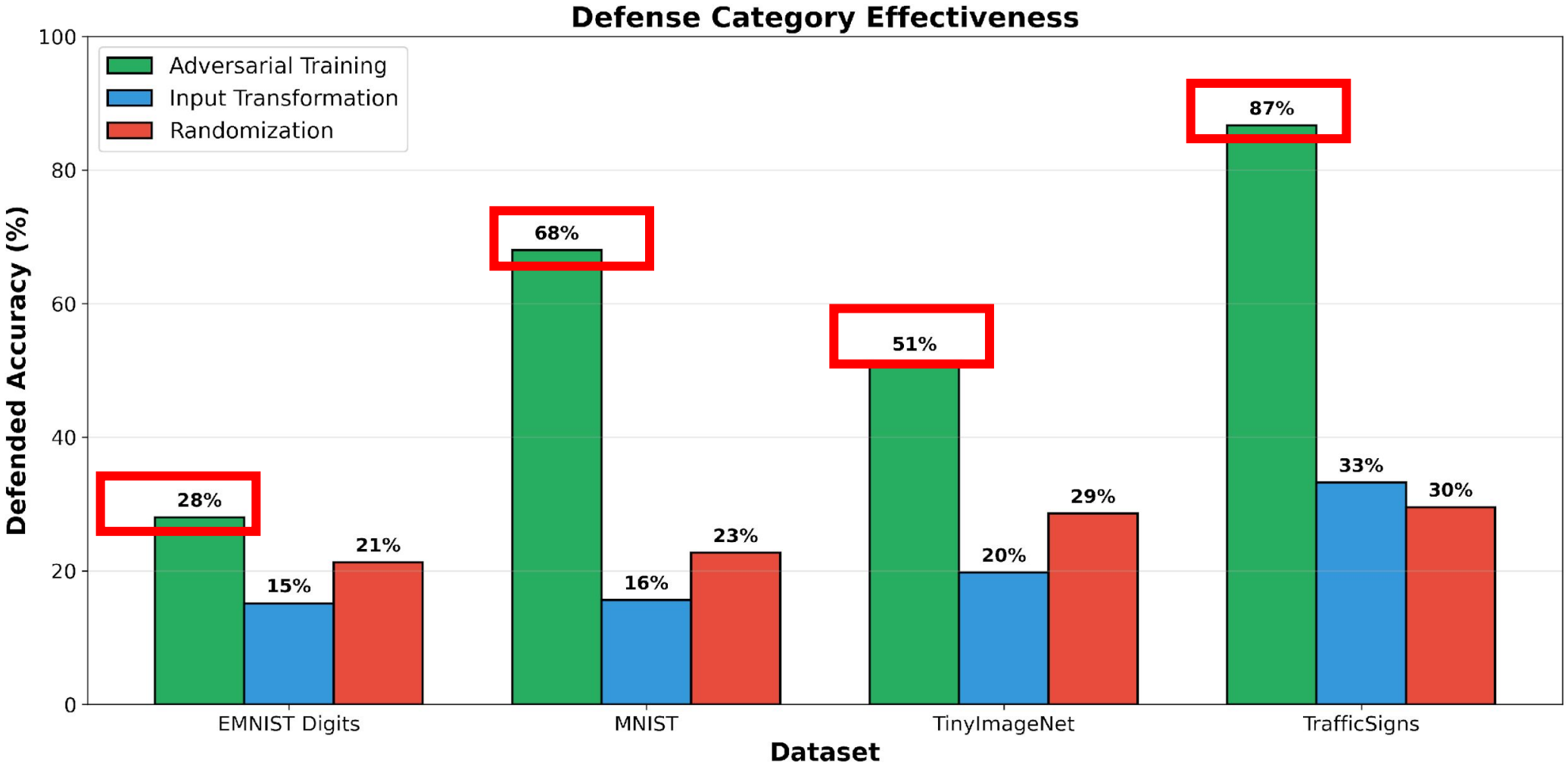
Preserved Accuracy = Defended Accuracy ÷ Clean Accuracy

Standardized metrics to quantify attack severity, defense recovery, and overall robustness improvement

Agenda

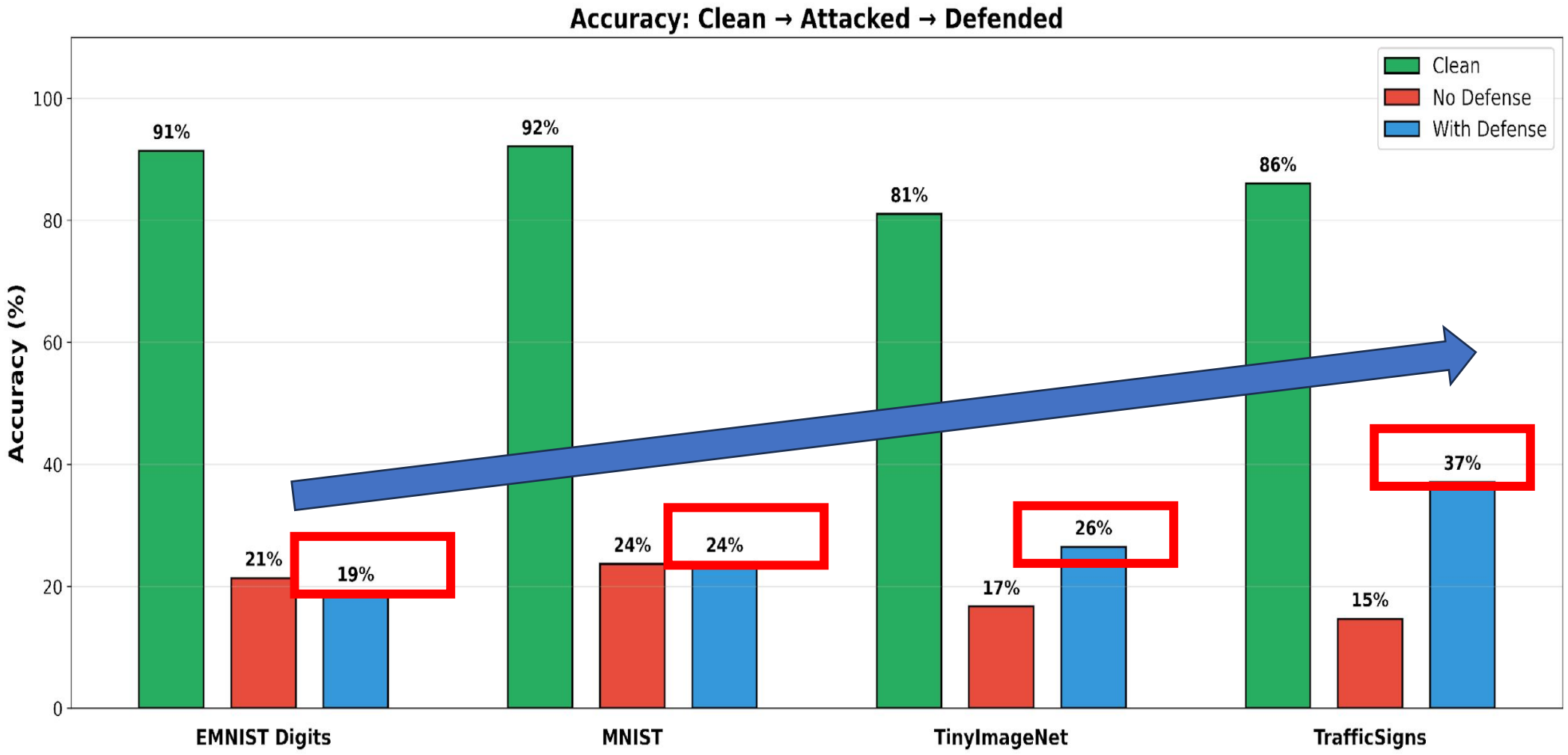
- Introduction
- Literature Review
- Methodology
- **Results**
- Discussion
- Conclusion

Results (1 of 8):



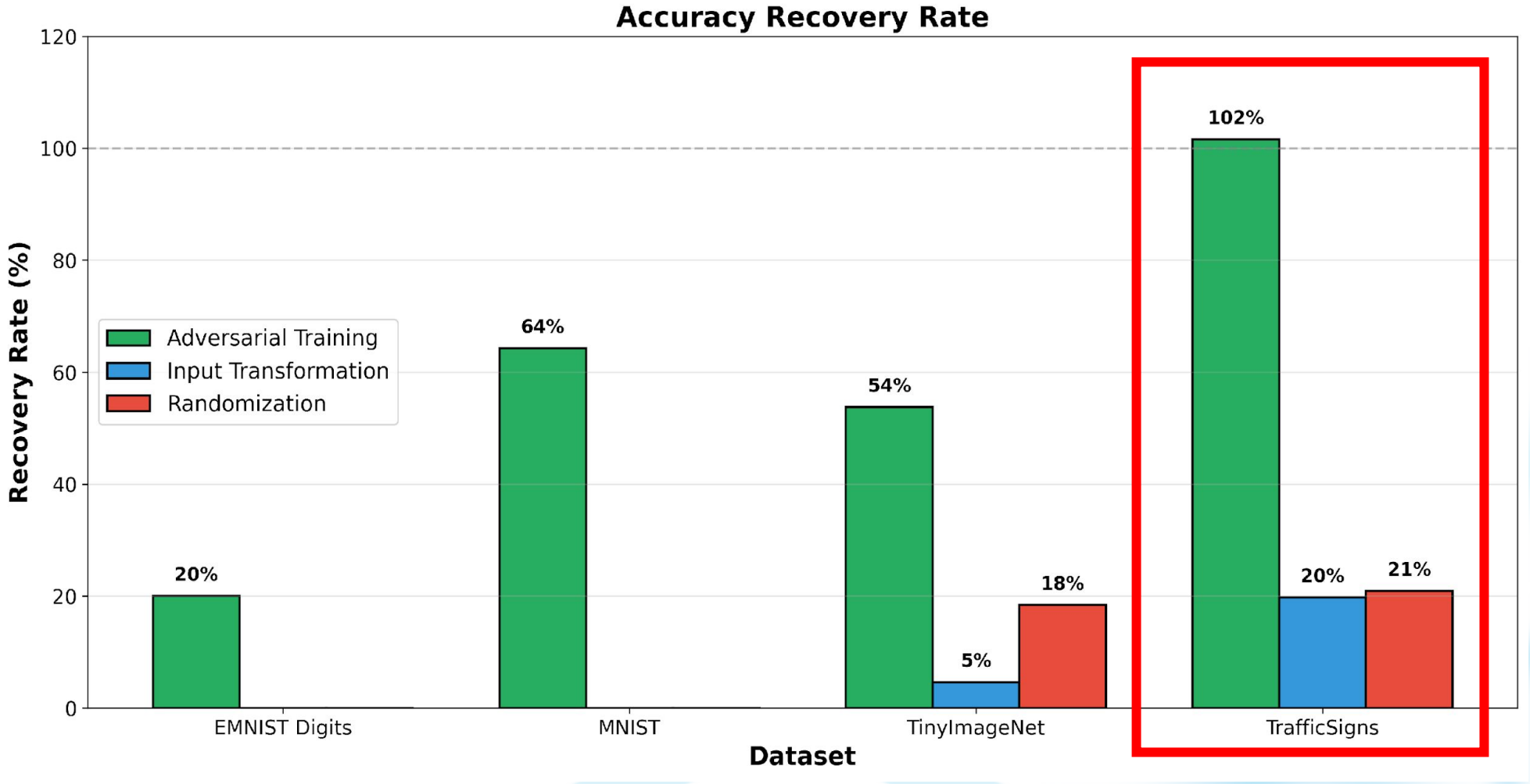
Adversarial training is the clear winner across all datasets.

Results (2 of 8):



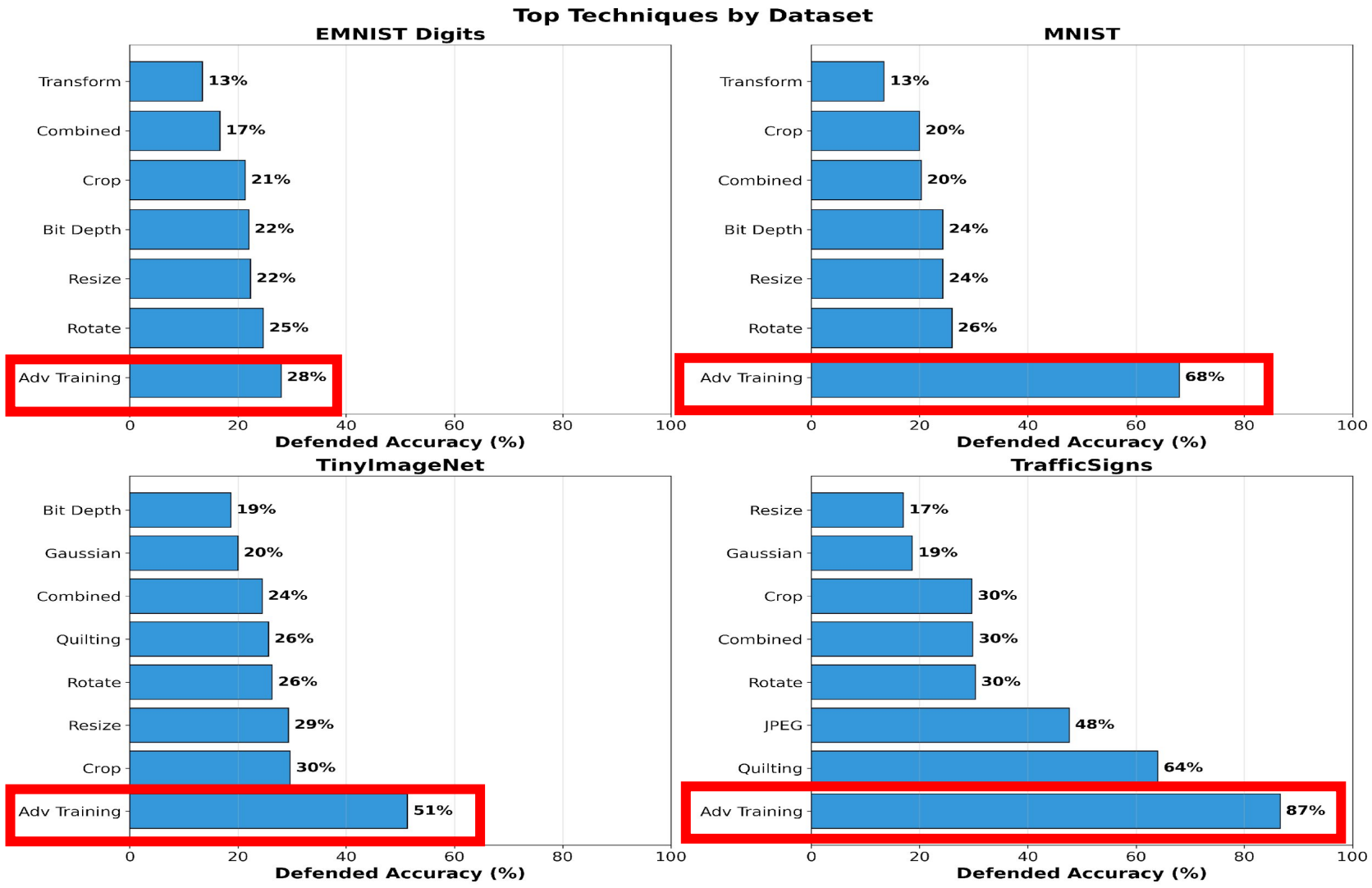
Adversarial training wins every dataset — no other defense comes close.

Results (3 of 8):



Adversarial training fully recovers TrafficSigns accuracy — exceeding baseline on structured data.

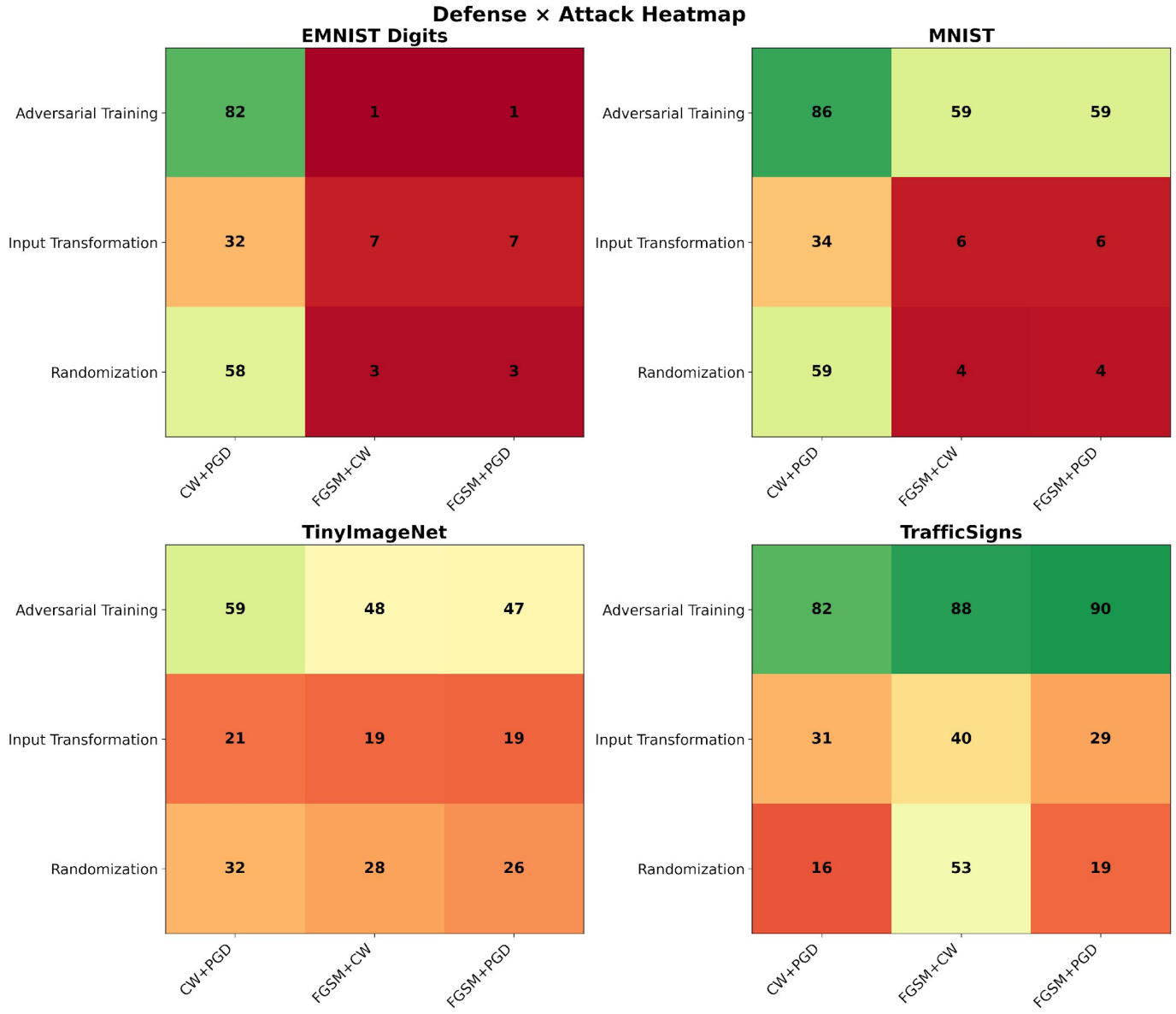
Results (4 of 8):



Adversarial training dominates — but dataset complexity determines how much.

Results (5 of 8):

- 80–90% — Excellent recovery
- 55–79% — Good recovery
- 40–54% — Moderate
- 20–39% — Weak recovery
- 1–19% — Defense failure

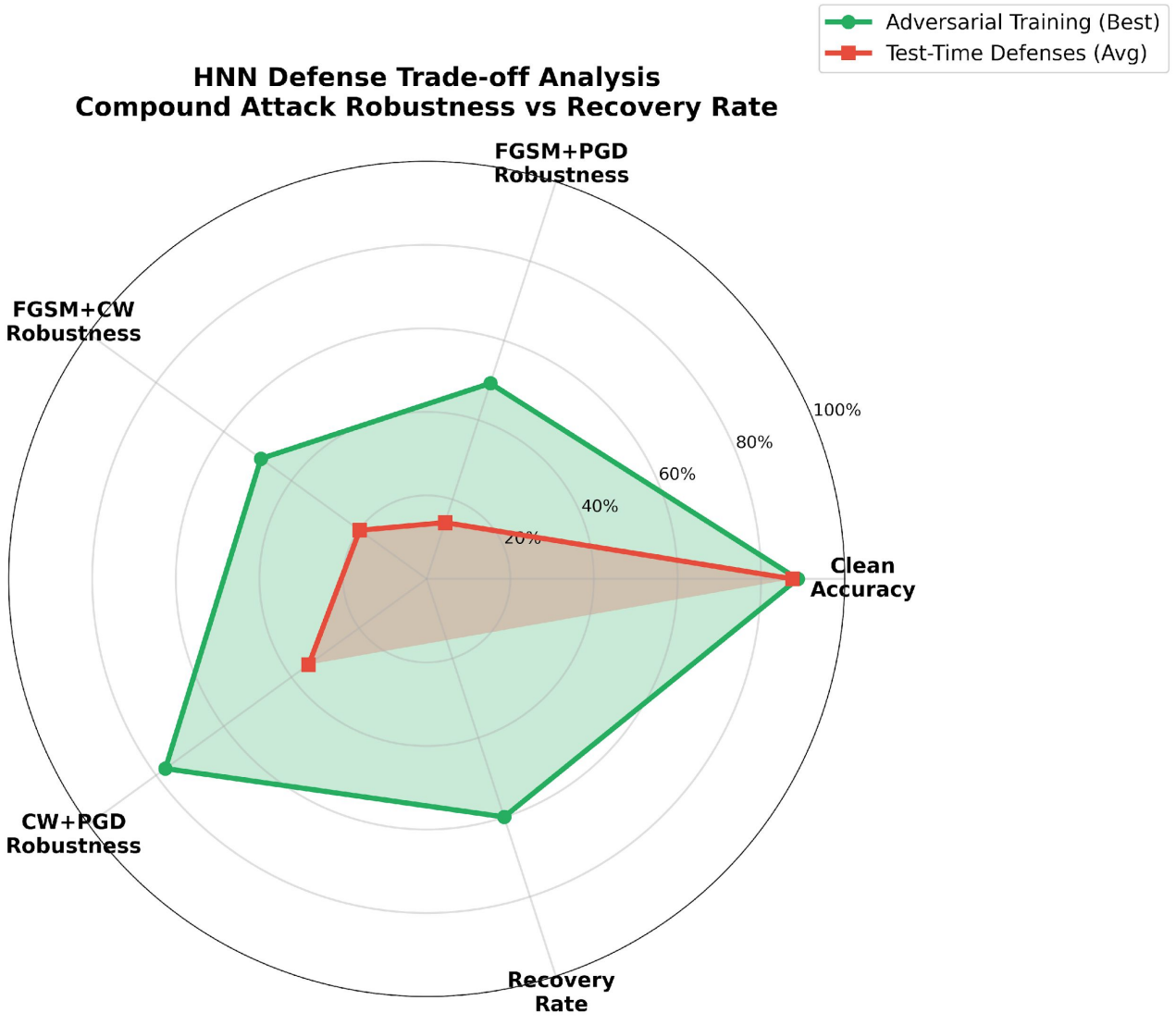


Attack type drives defense performance — no single test-time defense works across all attacks.

Results (6 of 8):

Compares two defense strategies (Training-time vs. Test-Time) across 5 performance dimensions.

A larger shaded area = stronger overall defense performance.

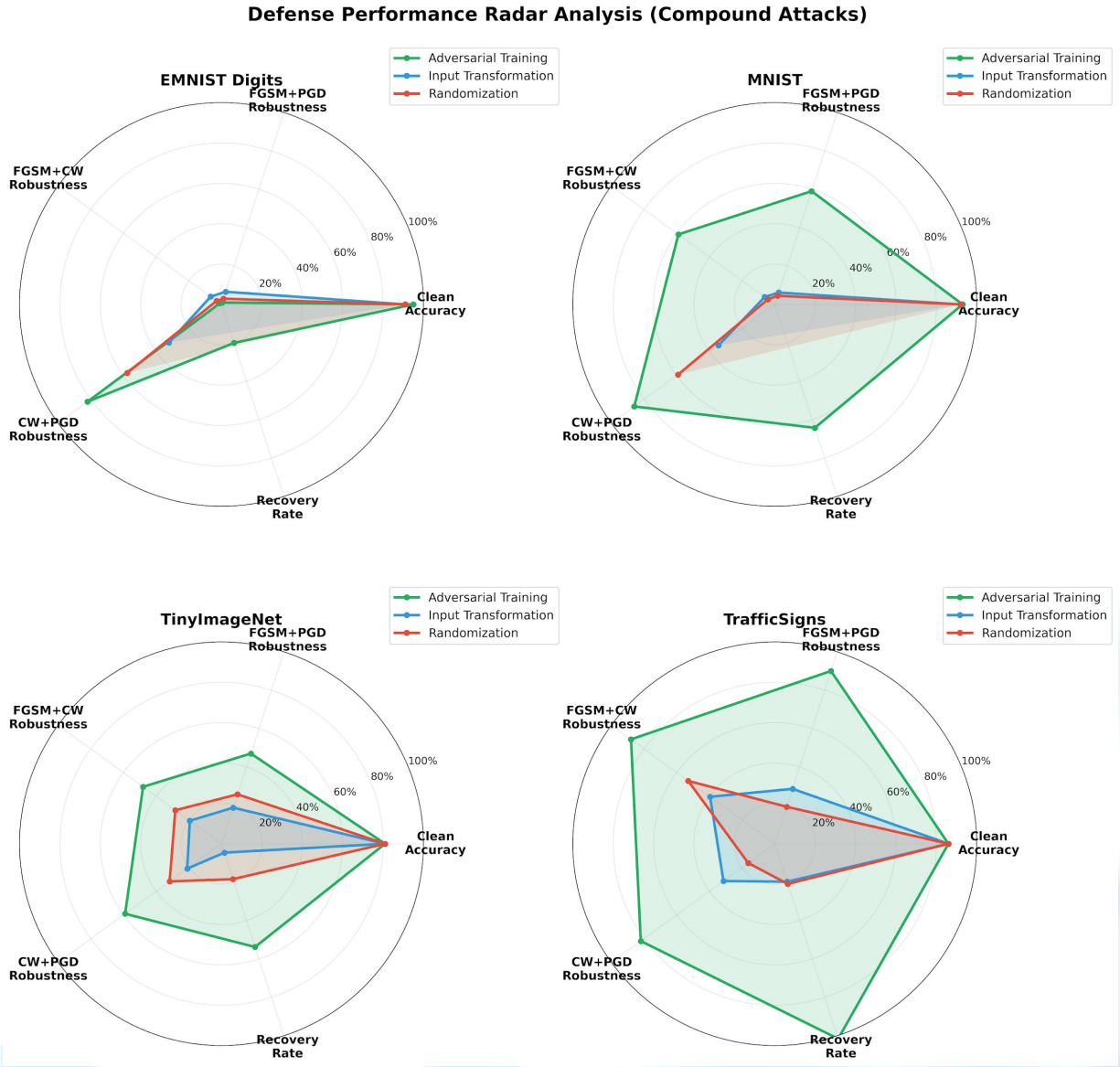


Higher robustness comes at the cost of deployment flexibility — adversarial training wins on both.

Results (7 of 8):

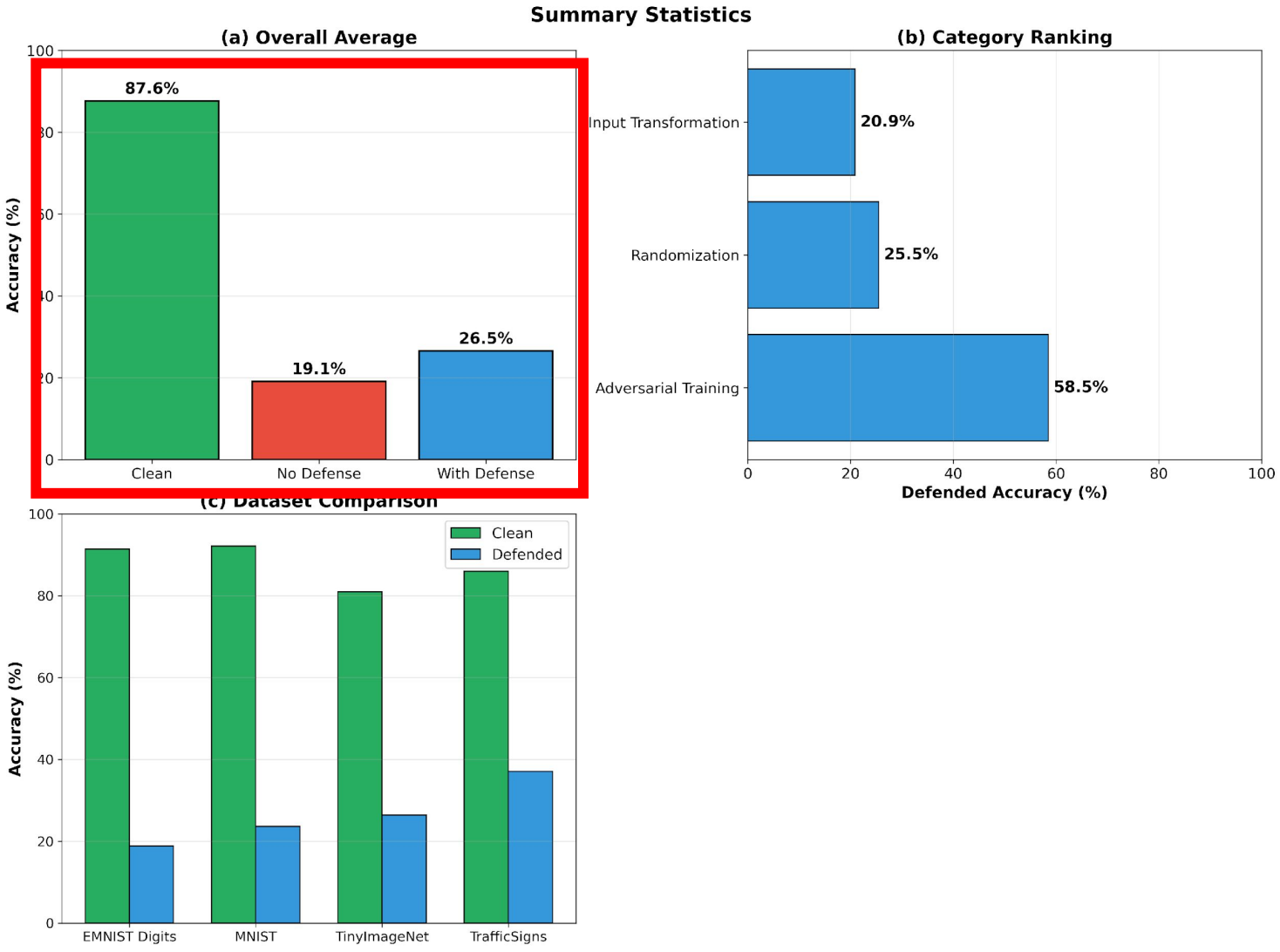
Compares three defense strategies (Training-time vs. Test-Time) across 5 performance dimensions for each dataset.

A larger shaded area = stronger overall defense performance



Compound attacks devastate HNN accuracy — only adversarial training reliably recovers it.

Results (8 of 8):



Adversarial training is the clear winner — all other defenses fall short.

Dissertation Contextualization

Quantum Adversarial Machine Learning Survey (Invited Springer Book Chapter)

- First systematic evaluation of multi-method compound attacks (FGSM+PGD, FGSM+CW, CW+PGD) against hybrid neural networks
- Novel quantum reservoir mapping approach contrasting with prior work using trainable quantum circuits
- Demonstrates effective defense prototyping using classical quantum circuit simulators before hardware deployment
- This work provides a pedagogical context for the dissertation

A survey paper comprehensively reviews adversarial threats and defense strategies for quantum and hybrid quantum-classical machine learning system.

Source: Quantum Adversarial Machine Learning and Defense Strategies: Challenges and Opportunities. *Springer Quantum Science and Technology Series*, in production.

Convergence Across 4 Case Studies

Key Results

- **Superiority of adversary training:** In all case studies, adversary training achieved 2.1-2.8× higher defended accuracy than test-time defenses.
- **Compound Attack Severity:** Compound attacks proved highly effective across diverse domains, achieving 73-84% attack success rates.
- **Geometric Priors Insufficient:** Architectural inductive biases improve clean accuracy but do not inherently provide adversarial robustness without explicit defense training.
- **Domain-Specific Challenges:** Each application domain introduces unique complications (class imbalance, environmental variation, safety certification requirements)
- **Certification Gap:** Defended accuracy (67-72%) falls short of safety-critical certification requirements (>95% for ISO 26262, DO-178C)

The four case studies demonstrate consistent patterns that reinforce the findings of the dissertation.

Case Study 1: Retroreflectivity-Based Traffic Sign Safety

Key Results

- Adversarial training achieved 70.98% defended accuracy
- HNN clean accuracy: 95.98% vs. classical CNN clean accuracy: 91.52% (4.46 per-centage point advantage)
- FGSM+PGD reduced the clean accuracy from 95.98% to 18.75% without defense(80.5% attack success rate)
- Consistent with the dissertation's finding of an advantage of 2.3-2.8× advantage over test-time defenses

The retroreflectivity classification of traffic signs represents a safety-critical application.

Source: Reflectivity-Driven Traffic Sign Safety with Synthetic Data and Defense Models. *IEEE TITS*, under peer review (T-ITS-25-08-4197).

Dataset: Ruganuza, D., et al. Creating a Comprehensive Dataset to Explore Retroreflectivity Degradation in Traffic Signs: Incorporating Environmental Factors. SC EPSCoR State Conference, Columbia, SC, April 2024.

Case Study 2: Cross-Domain Adversarial Robustness Analysis

Key Results

- Classical CNN performance achieved ¿95% clean accuracy across all three domains
- Geometric constraint networks showed 0-64% defended accuracy, indicating structural priors provide limited adversarial robustness
- Adversarial training defenses developed for traffic signs transferred effectively to lane detection (similar geometric structure) but poorly to crop detection (different structural patterns)
- None of the evaluated architectures achieved the > 95% robustness required for ISO26262 or DO-178C certification

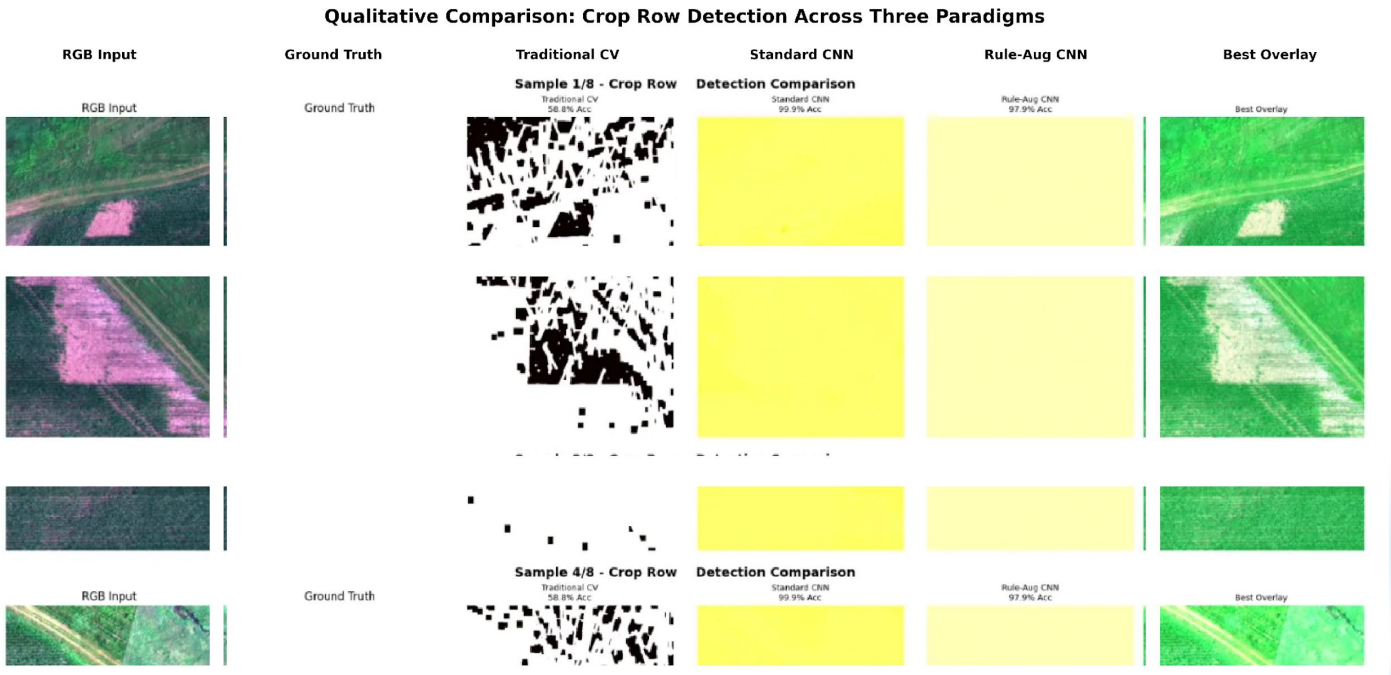
Cross-domain robustness analysis evaluates whether defense mechanisms transfer across distinct safety-critical applications.

Source: Cross-Domain Adversarial Robustness of Geometric Prior Neural Networks: Safety-Critical Deployment Guidelines. IEEE Transactions on Emerging Topics in Computational Intelligence, under peer review (TETCI-2026-0503).

Case Study 3: UAV-Based Crop Row Detection Protection

Key Results

- HNN achieved 93.2% clean accuracy
- Compound attacks reduced accuracy to 12-18% without defense
- Adversarial training restored the accuracy to 67.4%
- Input transformation defenses performed poorly (22-28%) due to natural variability resembling adversarial perturbations



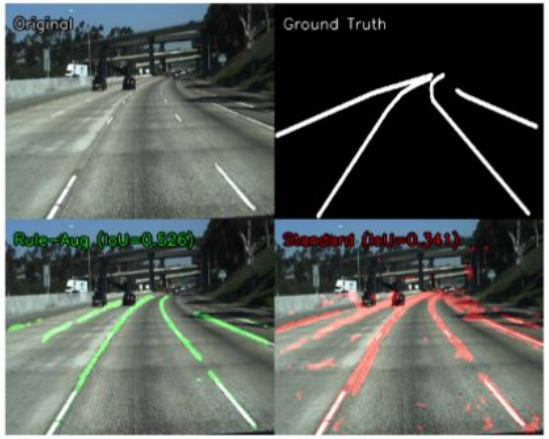
UAV-based agricultural monitoring relies on accurate crop row detection for automated harvesting, pesticide application, and yield estimation.

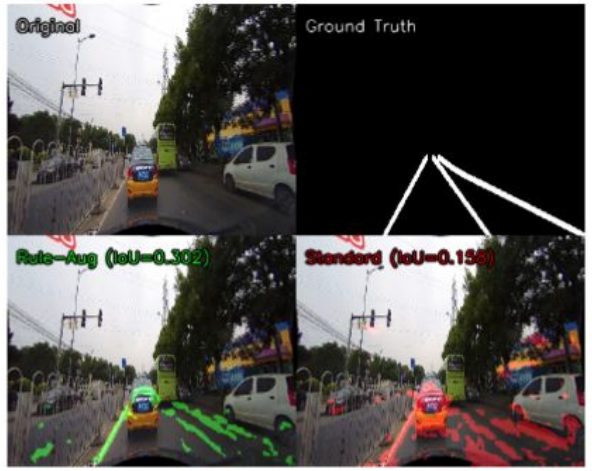
Source: Protection of UAV-Based Crop Row Detection Against Adversarial Attacks. IEEE Transactions on Emerging Topics in Computational Intelligence, under peer review (TETCI-2025-2766).

Case Study 4: Lane Detection Architectural Robustness

Key Results

- HNN with lane constraints achieved 96.8% clean accuracy vs. CNN 92.3%
- Geometric priors provided minimal defense (34% defended accuracy) without explicit adversarial training
- Combined geometric priors + adversarial training achieved 72.1% defended accuracy
- 72.1% falls short of the ISO 26262 requirement (>95%), indicating the need for ensemble or certified defenses





Autonomous vehicle lane detection is a safety-critical function where adversarial attacks could cause lane departure incidents or navigation failures.

Source: Architectural Robustness in Lane Detection Systems: Geometric Constraints and Adversarial Stability. Transportation Research Interdisciplinary Perspectives (Elsevier), under peer review (TRIP-D-26-00418).

Agenda

- Introduction
- Literature Review
- Methodology
- Results
- **Discussion**
- Conclusion

Mapping Hypotheses to Research Questions.

Hypothesis 1 & 2 → Answer RQ1 **(Effectiveness)**

- **H1:** HNN without defense = unacceptable accuracy under WTC attacks
- **H2:** HNN with defense = acceptable accuracy under WTC attacks

Together H1 & H2 test RQ1: *By comparing models with vs. without defenses, demonstrating whether defenses are effective*

Hypothesis 3 → Answers RQ2 **(Optimal Defense)**

- **H3:** At least one optimal defense will maintain highest acceptable accuracy

H3 tests RQ2: *Identifies which specific defense is most effective*

Research Question [RQ1]

RQ1: How effective are defense mechanisms at protecting HNN models against white-box, targeted, and compound (WTC) adversarial attacks?

- **Adversarial Training is Mandatory:** It is the only defense that consistently works, outperforming other categories by over 2x in accuracy.
- **WTC Attacks are Lethal:** Even with defenses, attacks like FGSM+PGD maintain a near-100% success rate on most datasets.
- **Low Recovery for Basics:** Simple techniques (Input Transformation/Randomization) are "weak" defenses, often failing to break 25% accuracy.

Research Questions [RQ2]

RQ2: Which defense mechanism is the most effective (optimal) for protecting HNN models against WTC adversarial attacks?

- **Complexity = Better Defense:** Defenses work much better on complex data (TrafficSigns) than on simple digits (EMNIST).
- **The "Ceiling" Effect:** Simple datasets have a performance cap; no matter the defense, accuracy rarely exceeds 30% on digits.
- **Robustness Scaling:** A model's ability to stay robust (Defended vs. Clean) triples as the imagery becomes more complex.
- **Specialized Success:** Advanced methods like Quilting only show their true value on high-complexity datasets.

Hypotheses [H1]

H1: An HNN model without any defense mechanism will exhibit unacceptable prediction accuracy when subjected to white-box targeted compound (WTC) adversarial attacks across multiple datasets, including MNIST, EMNIST "Digits", TinyImageNet, and TrafficSigns.

- **Hypothesis Supported:** HNN models without defense exhibit unacceptable accuracy under WTC attacks, with performance crashing from ~88% (Clean) to just 19% (Attacked) on average.

- **Total Collapse:** Attack success rates reach near 100% on three out of four datasets when no defense is present.

Hypotheses [H2]

H2: An HNN model incorporating defense mechanisms will achieve acceptable prediction accuracy following a WTC adversarial attack, using the same set of datasets as input.

- **Hypothesis Supported:** Every dataset tested contains at least one defense mechanism that increases accuracy above the undefended baseline.
- **Performance Gains:** Even in highly vulnerable datasets like EMNIST Digits, the top defense (Adversarial Training) improves accuracy to 28% from a 21% baseline.
- **Near-Total Recovery:** On the TrafficSigns dataset, the top-tier defense restores accuracy to 87%, virtually matching the model's original clean performance.

Hypotheses [H3]

H3: Among the defense strategies evaluated, at least one optimal defense mechanism will allow the HNN model to maintain the highest acceptable prediction accuracy under adversarial conditions of the WTC in all datasets.

- **Hypothesis Supported:** Adversarial Training is identified as the optimal defense mechanism across all datasets and attack types.

- **Superior Robustness:** It consistently maintains the highest acceptable accuracy, outperforming Randomization and Input Transformation by an average of 33% in defended accuracy.

Agenda

- Introduction
- Literature Review
- Methodology
- Results
- Discussion
- **Conclusion**

Conclusion

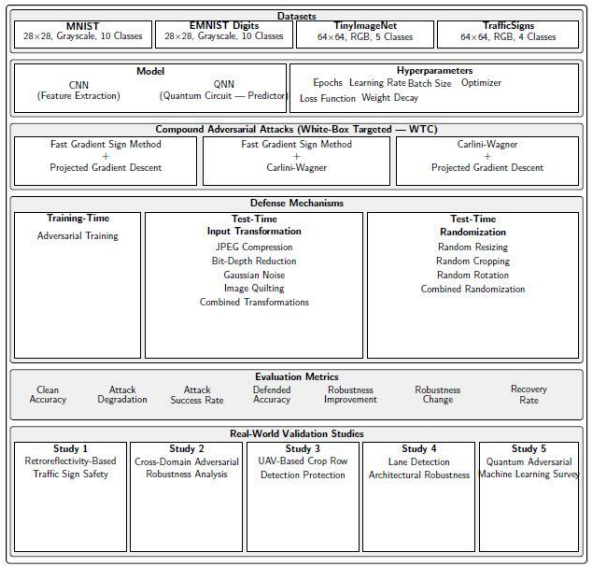
Key Findings

- **Compound attacks:** HNN accuracy collapsed from 87.6% to 19.1%
- **Adversarial training:** only defense with consistent recovery (28%–87%)
- **Dataset-dependent:** structured data (TrafficSigns) defends better than digit datasets
- **Classical simulation:** sufficient for HNN defense research without quantum hardware

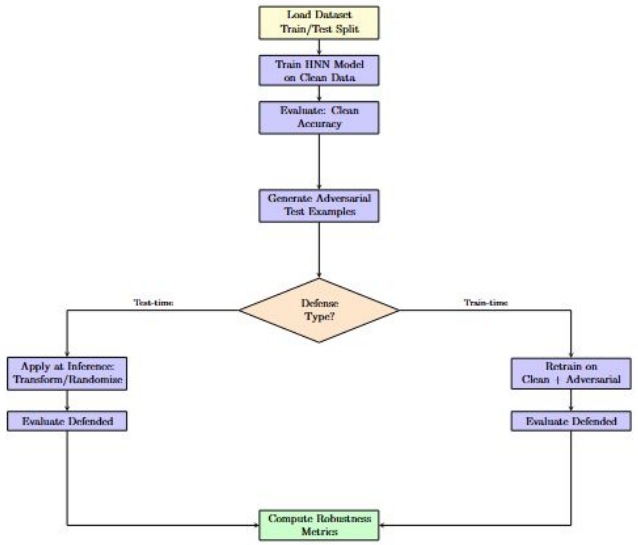
Future Work

- Validate classical simulation findings on physical quantum hardware
- Expand to adaptive and black-box attacks
- Extend to larger datasets and additional domains

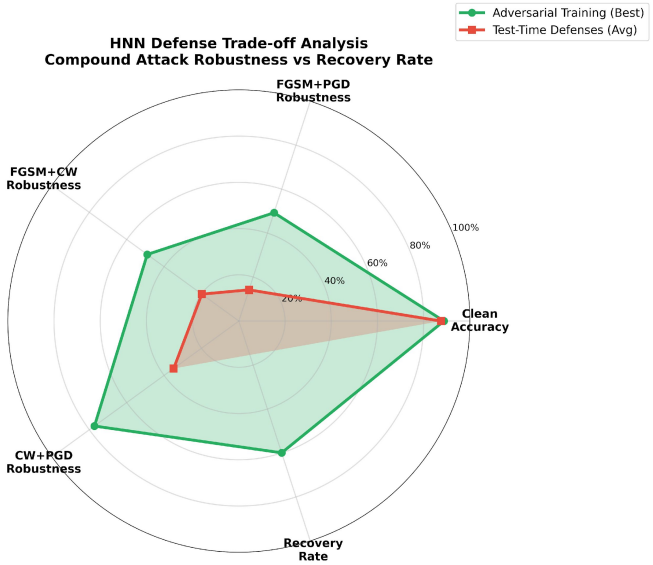
Goal: Protect HNNs from WTC attacks



Method: 10 defenses, 120 conditions



Results: Adversarial training dominates all



Findings: Dataset complexity drives effectiveness